A Smart Heating Set Point Scheduler Using an Artificial Neural Network and Genetic Algorithm

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***Abstract*—This paper introduces a novel, adaptive, heating set point scheduler that aims to minimise the heating energy consumption in a building while maintaining thermal comfort levels. The presented control strategy couples two computational intelligence techniques, an Artificial Neural Network, ANN and a Genetic Algorithm, GA. The ANN surrogate model was trained using data from multiple Energy Plus building simulations with varied heating set points. This allows quick, computationally cheap, calculation of a fitness function, opposed to time consuming Energy Plus simulations. The ANN's inputs include weather and occupancy predictions as well as taking account of the buildings thermal inertia to predict energy consumption, predicted percentage dissatisfied, PPD, and indoor temperature. A multi-objective GA uses the ANN to calculate the energy sum over 24 hours and the average occupied PPD as its two objectives. The optimisation strategy was applied on a week in January over which it reduced energy consumption by 4.93% and improved PPD by 0.76%. The great advantage of this controller is that it could be simply adapted to take account of the changing energy environment. This could include an addition of renewable resources or demand response controls as part of a district heating network.**

Keywords—Smart Buildings; Artificial Intelligence; Building Control; HVAC; Energy;

I. INTRODUCTION

Buildings account for 40% of global energy consumption and contribute 30% of greenhouse gas emissions [1]. HVAC, (heating, ventilation and air conditioning) systems typically account for between 40% and 60% of total building energy consumption [2]. Therefore, the building sector is a key sector to target for efficiency savings to meet global targets of energy and emission reductions. It has been estimated that building energy consumption can be reduced by 20% – 30% by developing smarter, more optimal controllers without needing to change any hardware within the building [3]. This combined with retrofitting older buildings could contribute large savings from the building sector. Furthermore, a new generation of smart building controllers is needed to be able to cope with large shifts in the energy infrastructure. The increased use of small scale, decentralised, renewable energy generation such as solar photovoltaics (PV), solar thermal and wind power, mean that controllers will have to forecast and plan to maximise the use of these stochastic resources. The introduction of district level microgrids also put demands on building controllers as the controller not only has to manage its own building but also consider other buildings peak demands.

Current research into smart building controllers tends to focus on the use of model predictive control, MPC. MPC includes a predictive element that allows the buildings thermal mass to be utilised as well as any energy price variation throughout the day. It also re-optimises at relatively short time steps to take disturbances, such as weather or occupancy, into account as well as errors in previous predictions. The prediction model is an essential part of MPC however there is no consensus in the literature as to the most appropriate model. White box, physics based models such as energy plus are extremely detailed but are computationally expensive and cannot be used in online building control combined with meta-heuristic, searching algorithms [4]. Grey box models are reduced physics-based models such as Resistor-Capacitance, RC, models. With a small amount of data these can be calibrated to give a fairly accurate and quick prediction of building performance. Black box models based on statistical methods or learning methods such as artificial neural networks can also be used. These models are trained using large amounts of historical data with no knowledge of physical entities and once complete provide a very fast calculation. The disadvantages of black box models are the amount of training data needed and the fact they can only accurately predict the building conditions that they have been trained with [5].

An example of MPC being used to control HVAC systems can be found in [6]. In this work, they created an RC model of a Czech university building and the MPC strategy took into account the weather, price variations, comfort criteria and occupancy. This was trailed over a three-month period and achieved savings of 17% for a less insulated block and 27% for a well-insulated, refurbished block. Ma, [7], developed a control architecture that combines an Energy Plus model and an optimisation in MATLAB where the data exchange is facilitated by the building control virtual test bed, BCVTB, middleware. This gave energy savings of around 25% over rule based strategies although a thermal comfort index is not used. Oldewurtel, [8], makes an adaptation to basic MPC to create a stochastic MPC. This is different to the other approaches as it specifically considers the uncertainties in the forecasts it receives. This means the controller is slightly more conservative in its actions, it will not schedule activity that the model predicts are very close to the comfort boundary if there are large

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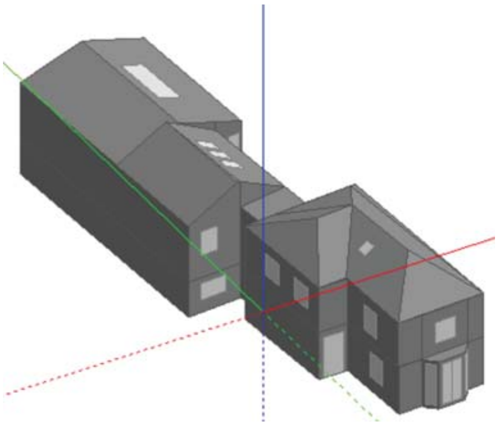

Fig. 1. Energy Model of the Office Building

uncertainties in the forecasts it receives. Yuce, [9], coupled an ANN and a GA to control smart appliances within a holiday home. Using this, grid energy reductions of 10%, 25% and 40% were imposed on the building and this in turn increased the utilisation of the local renewable resources by scheduling device activity around this production.

This paper will introduce a novel heating set point scheduler for an office building in Cardiff, UK. It combines an ANN model of the building with a GA, to minimise the energy consumption whilst maintaining thermal comfort over a 24-hour period.

II. CASE STUDY BUILDING DESCRIPTION

A. Energy Modelling

The building used for this case study is the office of the BRE Trust Centre for Sustainable Engineering in Cardiff, UK. The interior of the building was scanned using a Faro 3D laser scanner from which an as-built representation of the building could be formed in Autodesk Revit, a building information modelling, 3D design tool. From this model, floor plans, elevations and sections could be exported allowing the creation of an accurate energy simulation model in the Design Builder software package, as shown in Fig. 1. Known information such as the insulation levels of the walls, roof, floors and windows were assigned to the model appropriately and party walls with the adjoining building were set to adiabatic, as no heat transfer was assumed. The building contains 6 conditioned zones with around 30 occupants in total. It includes 3 office spaces, a kitchen, a meeting room and a reception. The building is naturally ventilated and cooled in the summer and uses a typical gas boiler and hot water, radiator heating system. Typical occupancy, equipment and lighting schedules were inputted to the model based on occupant surveys. These settings have an impact on the heating load and consequently, on energy consumption, within the building.

B. ANN Creation

An ANN is a black box, machine-learning method that it inspired by the way the human brain works. They are adept at handling highly complex, non-linear problems with noisy data [10]. However, they require large training data sets to learn the characteristics and trends in the data. The ANN receives the inputs and the corresponding outputs from which a training algorithm can set the internal weights and biases within the model to provide the best prediction of the given data set. Once this training is complete, the weights and biases are fixed and only inputs need to be provided for the ANN to calculate the chosen outputs. The simulation model described in Section II-A, was used to produce the large volumes of data necessary to train an accurate ANN. The resulting surrogate model is able to fulfil the role of the building simulation model with massively reduced computational requirements. The period under consideration was from the 1st of January to the 31st of March as heating is required over this period. A number of simulations were run with different heating set point schedules. These schedules ranged from typical schedules to ones where the set point changed randomly between 12℃ and 24℃ every hour.

Once the initial training was complete the optimisation was run. The results of the initial optimisation gave solutions where the ANN underpredicted the energy consumption and PPD compared to an Energy Plus model with the same schedule. So, the first optimisations were effectively used to identify some of the gaps in the ANN training data. To improve the ANN to acceptable standards these initial 'optimal' schedules were fed back and used as training schedules in an iterative procedure. This breadth of simulations aims to give the ANN some understanding of all possible scenarios. The resultant input and output data of the 27 independent simulations were compiled into a single training data set which held 58237 hours of diverse data, something which can realistically only be achieved using simulation models.

Fig. 2 provides a diagram of the ANN inputs and outputs. In our case study the inputs needed to be variables that could be obtained 24 hours in advance to allow this strategy to be

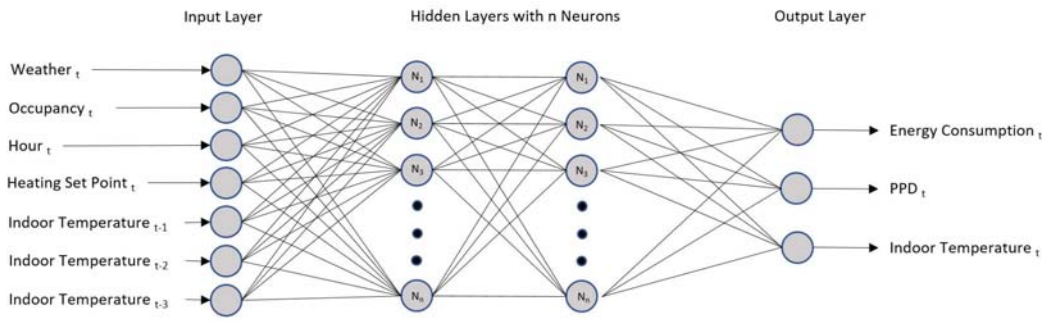

Fig. 2. ANN Inputs and Outputs

deployed in real time. Therefore, weather inputs of outdoor temperature, solar irradiance and humidity were chosen as these are widely available from local weather stations. Hour of the day and an occupancy profile were also chosen as inputs. Note that for occupancy the controller would simply decide whether it is a typical working day and if so a typical occupancy pattern is the input. This is due to the difficulty in predicting and measuring precise occupancy. As well as these uncontrolled variables, the decision variable, the heating set point temperature was also an input. To capture the heating inertia effect, caused by a buildings thermal mass, the indoor temperature for the previous 3 time steps were also used as inputs. The outputs needed for the GA optimisation algorithm were the energy consumption and the predicted percentage dissatisfied, PPD, a thermal comfort measure. However as indoor temperature is also an input this also needs to be predicted by the ANN to provide a 24-hour forecast. Therefore, indoor temperature for the next time step is another output but not an optimisation objective.

The ANN was created using the MATLAB ANN toolbox. ANN have several parameters that can be tuned and adjusted which affect the quality of the output prediction. These include the number of hidden layers, number of neurons in each layer, the transfer function and the training function. There are no simple rules to follow to set these parameters, they need to be iteratively tuned to find the best architecture. In this study we have used the same methodology outlined in [11] and the resulting architecture has 2 hidden layers, 20 neurons in each layer, Log-sigmoid transfer function and Bayesian regularization backpropagation as a training function. Throughout the ANN generation, little difference was found between Bayesian regularisation and Levenberg-Marquardt training functions or Log-sigmoid and Tan-sigmoid transfer functions. The most sensitive parameter was the number of layers and the number of hidden neurons. Having 2 hidden layers markedly improved prediction accuracy as did increasing the number of hidden neurons. However, eventually the improvement plateaued and the training time increased so 20 neurons in each layer was the final compromise. The resulting final ANN had a Pearson correlation, a measure of linear correlation between target value and prediction, of 0.9888 for energy consumption, 0.9982 for PPD and 0.9985 for indoor temperature predictions. The prediction and target values for energy consumption, PPD, indoor temperature prediction as well as set point for a sample week are shown in Fig. 3. This demonstrates excellent prediction despite a very challenging set point schedule.

III. Optimisation Strategy

The proposed control strategy aims to minimise the heating energy consumption from the boiler whilst maintaining good thermal comfort within the building. Building temperature control is a highly non-linear problem so simple optimisation techniques are not appropriate to solve building control problems [12]. Therefore, this strategy will use a genetic algorithm, GA, which have been shown to work well with highly non-linear building behaviour [13]. The GA will be coupled with the ANN which will provide the building modelling. In this study, we have implemented the multi-objective GA in the MATLAB Optimisation toolbox.

GAs are population based, iterative, searching algorithms inspired by the process of natural selection. It starts with a randomly generated population of solutions. Each of these solutions in our case contain 24 values bounded between 12℃ and 24℃ representing the heating set point every hour for 24 hours. Each of these 24-hour solutions is passed through the ANN developed in Section II. From this, the sum of the energy consumption over the 24 hours is calculated, as well as the

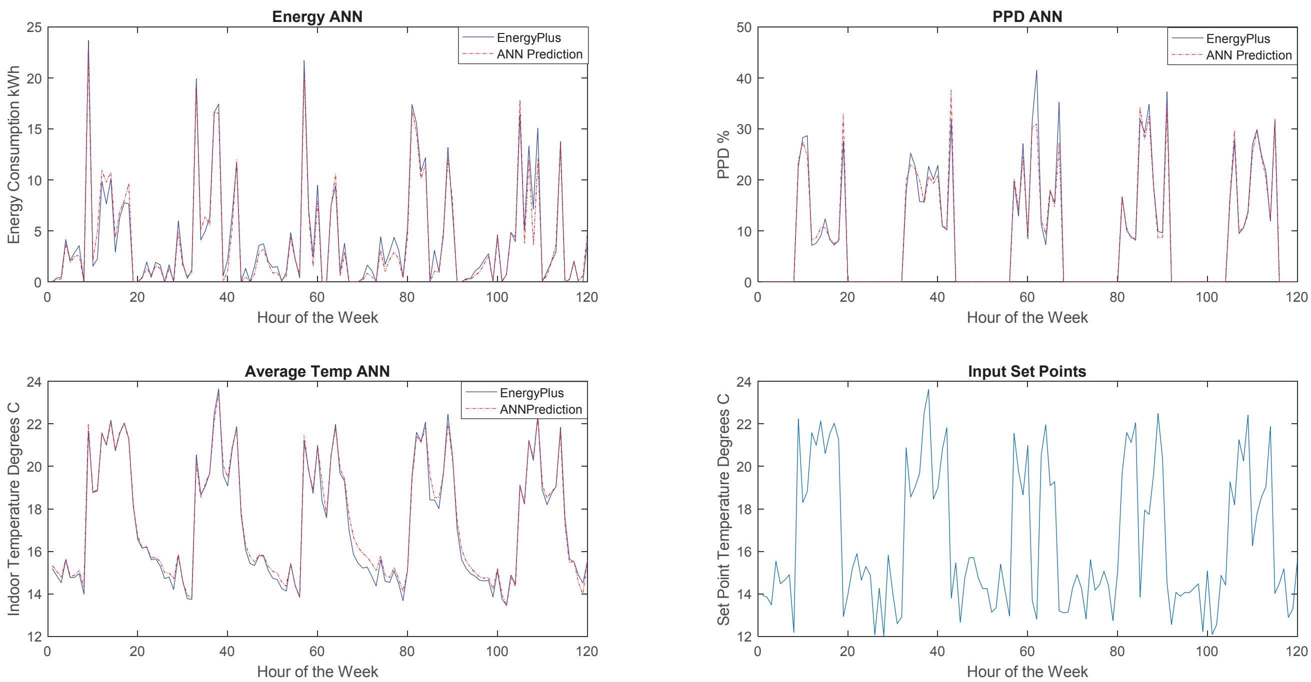

Fig. 3. ANN Prediction vs Energy Plus Prediction

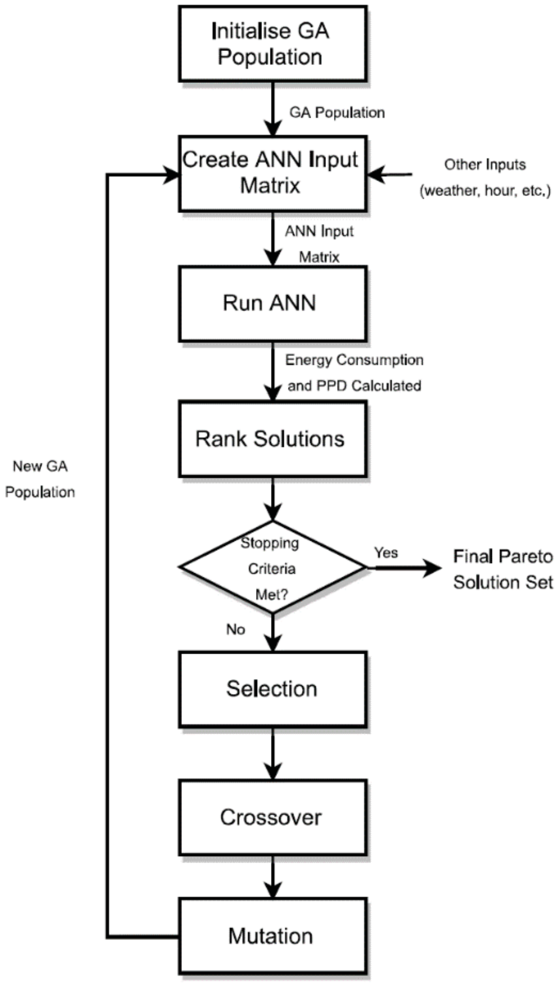

Fig. 4. Optimisation Procedure

average PPD for the occupied hours. These are the 2 objectives in the fitness function. The individual solutions are then ranked based on their fitness which influences their chances of being selected to reproduce the next generation of solutions, analogous to Darwin's 'survival of the fittest'. Two selected 'parent' solutions are then crossed over to produce a 'child' solution with combined genes. Then, the mutation stage can cause some of the genes to vary by a selected function to ensure diversity of solutions so the algorithm does not get stuck in a local minimum. After this, an entirely new generation of solutions is formed and the process repeats until stopping criteria are met, this is displayed in the flowchart in Fig. 4.

As the PPD objective is an average this means the PPD at one hour can be high but overall the average could still be competitive. Obviously, this is unacceptable in practice so to address this we have imposed an additional penalty function at an hourly level. If the PPD goes over 20% at any one hour, then it is replaced with:

$$PPD = PPD + (PPD - 20)^2 \quad (1)$$

This will strongly discourage solutions that have very high PPD for just a single hour. As this is a multi-objective algorithm, once completed a Pareto Front of 100 non-dominated solutions are produced. Theoretically, none of the produced solutions are more optimal that the others but for this strategy to be implemented in real time an automatic way of selecting a solution from the set must be found. Our approach to this problem is to first run a baseline scenario through the ANN. This scenario is a typical building schedule where the set point is 12°C for all unoccupied hours and 21°C for the occupied hours. This gives us a baseline energy consumption and PPD as a reference. We then filter the provided solutions by removing any solutions where either the PPD average or sum of energy consumption is above 120% of the baseline scenario. This produces a filtered Pareto Front which is then normalised. The distances from each normalised point to the origin is calculated and the solution with the minimum distance is selected as the chosen solution. This means that the extreme solutions are removed automatically and only sensible solutions remain. This is demonstrated in Fig. 5.

IV. Optimisation Results and Discussion

The first full working week of the year without holidays, the 11th to the 15th of January, was chosen to be studied. Note that weekends are not considered by our optimisation strategy and the current setback temperature of 12°C is used. For each day 5 independent GA runs were undertaken to assess the consistency of the optimisation results. For the sake of brevity only results from one day, the 13th, are shown in Table 1. Each optimal solution found from the GA was put back into Energy Plus to validate the result. This removes the possible impact of errors in the ANN prediction and allows the percentage error between ANN and Energy Plus predictions to be calculated.

The overall average error in daily energy sum prediction between the simulation model, viewed as the target, and the ANN is 5.87% for 25 solutions over the whole week. The average error in PPD prediction over the same period is 3.42%. Throughout the independent runs on the same day similar

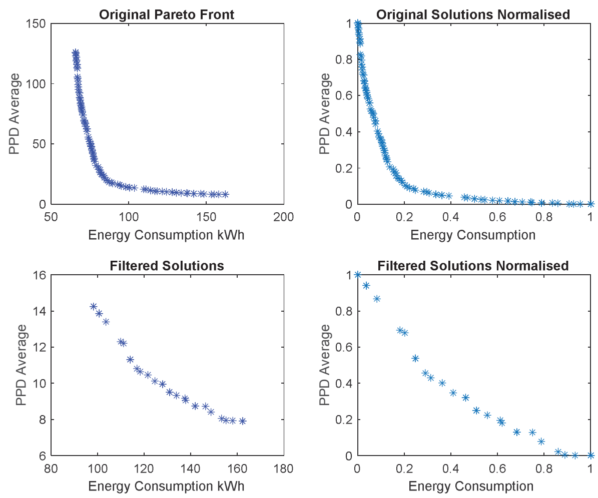

Fig. 5. Solution Filtering Example

TABLE I. FIVE INDEPENDENT RUNS OF THE GA ON JAN 13TH

Run Number	ANN Prediction		Energy Plus Calculation		% Error	
	Energy	***PPD***	***Energy***	***PPD***	***Energy***	***PPD***
Baseline	-	-	107.8	9.6	-	-
Run 1	99.35	9.94	105.86	10.16	6.15	2.13
Run 2	103.17	9.28	109.28	9.43	5.59	1.57
Run 3	102.70	9.63	105.42	9.88	2.58	2.48
Run 4	105.45	9.13	107.50	9.16	1.90	0.34
Run 5	111.29	9.54	111.62	9.35	0.30	2.02

solutions are found with little variation in the energy or PPD prediction over a single day. The variation that does exist is likely down to the filtering of the Pareto solutions displayed in Fig. 5. As this only considers a small section of the Pareto Front the point density and typical Pareto curve is lost and what remains is closer to a straight line. This means that very slight changes in point position can radically alter which solution is selected.

Interestingly if the building is particularly cold, like in the case of Monday 11th of January where the building had been held at the setback temperature of 12℃ for 2 days, the optimal heating strategy is to employ some pre-heating in the early hours. Whilst this causes an increase in energy consumption at that time it does give savings over the course of the day. In the case of Fig. 6 energy savings were 11.67% and average occupied PPD improved by 4.93% over the baseline scenario when comparing simulation against simulation. Also, noticeable in Fig. 6, the optimal solutions tend to have a more gradual increase in set point temperature in the morning and starts to reduce the set point earlier in the evenings compared to the baseline. This results in energy savings and is the case for almost every result across all five days. Reductions in energy consumption and average PPD are much smaller over the following four days. This indicates that the baseline scenario is very close to being optimal when the building begins the day at a moderate temperature (15°C-16°C). This could be due to the case study building being relatively poorly insulated and not very airtight. This means that pre-heating strategies, apart from in extremely cold circumstances, are not preferable as the building would not retain that heat energy. In future work, we aim to alter the simulation to make the building more airtight and insulated and re-run the optimisation to assess this hypothesis.

In real deployment, the GA would only be run once per hour, Table 1 is simply to assess the consistency of results. Therefore, one schedule from the 5 optimisations on each day was selected and compiled into a week-long optimised schedule. This custom schedule was then inputted into the simulation model. This can be compared to the baseline simulation where the set point is 12℃ for unoccupied periods and 21℃ for occupied periods. This allows a direct comparison as every aspect, including the weather, occupancy and equipment patterns, of the two simulation models are identical apart from the heating schedule. The results are shown in Table 2.

Over the course of this week the optimised control schedule gives a 4.93% energy saving with a 0.76% improvement in average PPD. Whilst this may be a relatively modest energy saving, the developed control strategy is far more adaptable than traditional rule based control. For example, if local renewable heat generation was available then this could be easily factored in to the control strategy allowing better utilisation of the available resources that traditional rule based control. Or one can envisage a scenario where this building was part of a wider

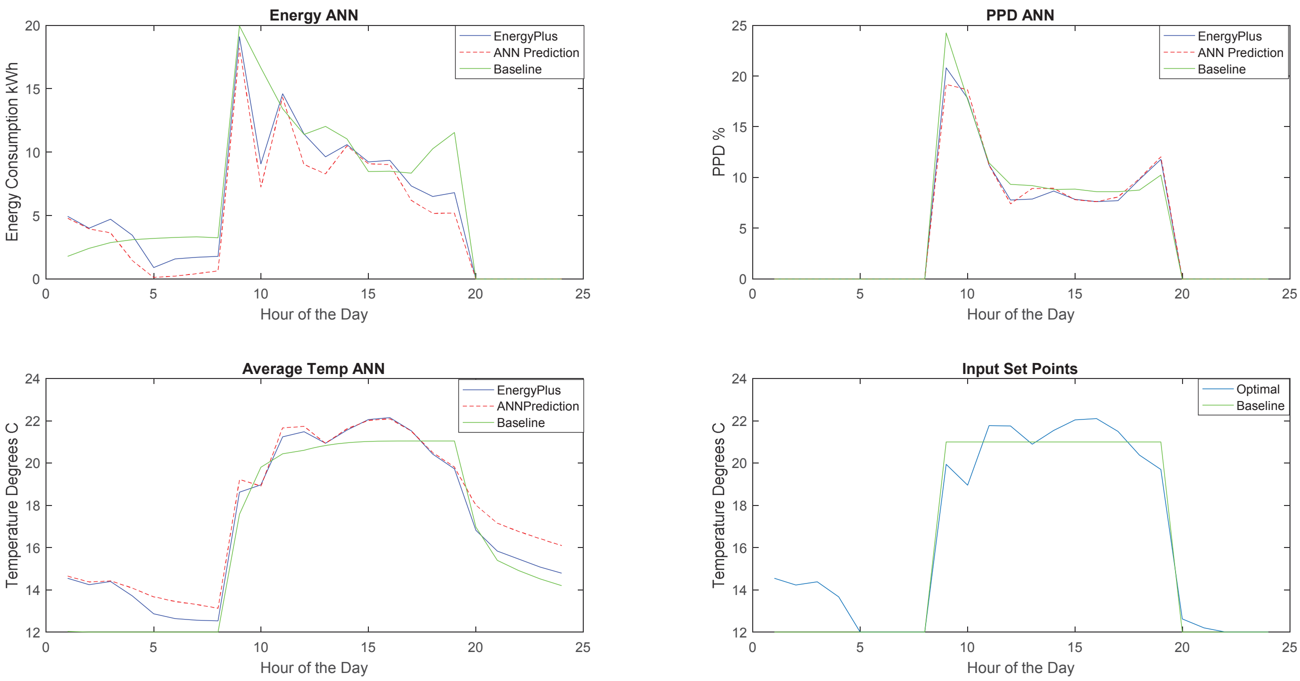

Fig. 6. January 11th Results

TABLE II. WEEK LONG OPTIMISATION RESULTS

	Energy Consumption		Average PPD		Energy Change vs Baseline %		PPD Change vs Baseline %	
	ANN	*E+*	*ANN*	*E+*	*ANN*	*E+*	*ANN*	*E+*
Baseline	-	621.36	-	10.34	-	-	-	-
Optimal	538.37	590.72	10.03	10.26	13.36	4.93	3.00	0.76

building cluster supplied by a district heating system. It may then be advantageous to impose demand response signals or time of use price tariffs. An adaption of this control strategy would be able to react to pricing changes to bring greater savings than fixed, rule-based controllers. Due to the surrogate model, this control strategy could be simply deployed with only the addition of temperature sensors in each zone and a remote link to the buildings' thermostat. When deployed, this strategy could easily work as MPC, utilising the sensed temperature feedback, and reoptimizing every hour to react to actual conditions and unforeseen disturbances.

Whilst the results found in this case study do not reach the 20%-30% theorised in the introduction, several adaptations will be made to the controller to gain greater savings. The building used in the case study is not well insulated, has reasonably high outdoor air infiltration and has single glazed windows. The authors believe that if the control strategy was applied to a more tightly insulated building then the potential for savings would be greater as pre-heating would be more effective. This is supported by the results found in [6]. Therefore, future work will be to apply this control strategy to a larger, more complex, and better insulated building and to implement this as real-time MPC. Furthermore, this work can be extended by optimising at a lower level to provide zone or room level optimised schedules. This would be able to take advantage of room occupancy profiles to only heat the rooms when required.

V. CONCLUSION

This paper presents a smart, heating set point scheduler with the dual objectives of minimising energy consumption whilst maintaining good indoor thermal comfort. To achieve this a small office building was modelled in Energy Plus and used to produce large volumes of data with varied heating set points. This data was used to train an ANN with weather information, set point, hour of the day, occupancy and previous indoor temperatures as inputs. This then accurately predicted the energy consumption, PPD and indoor temperature for the next 24 hours. The ANN was then used in a GA optimisation procedure to calculate the two objective values, the sum of the energy consumption over 24 hours and the average PPD for the occupied period. The Pareto front of solutions was filtered by removing the outer extreme solutions and the solution with the minimum distance to the origin when normalised was chosen as the selected optimal.

In a case study, a single week in January was optimised. Each day of the week was run 5 times independently. Each of the produced optimal schedules was then put back into the simulation model to a) assess the accuracy of the ANN against the simulation it is trying to replace and b) validate the solution against a simulation baseline scenario. Over the 25 runs the average error in energy prediction was 6.28% and for PPD was 3.27%. One solution was chosen for each day and combined into an optimal weekly schedule. This was run in the Energy Plus simulation model and compared to the baseline scenario. The optimisation achieved reductions of 4.93% in energy consumption and 0.76% in PPD. Whilst these are relatively modest improvements, this control strategy could be simply deployed as MPC with minimal extra equipment. Future work will assess the effect building insulation has on the energy savings and the control strategy will be further improved to consider zone level occupancy so unoccupied rooms are not heated.

ACKNOWLEDGMENT

The authors would like to acknowledge the financial support of EPSRC (Engineering and Physical Sciences Research Council) and BRE (Building Research Establishment).

REFERENCES

[1] M. W. Ahmad, J. Hippolyte, J. Reynolds, M. Mourshed, and Y. Rezgui, "Optimal scheduling strategy for enhancing IAQ , thermal comfort and visual using a genetic algorithm," in *IAQ 2016 Defining Indoor Air Quality: Policy, Standards, and Practices*, 2016.

[2] L. Pérez-Lombard, J. Ortiz, and C. Pout, "A review on buildings energy consumption information," *Energy Build.*, vol. 40, no. 3, pp. 394–398, 2008.

[3] X. Guan, Z. Xu, and Q. Jia, "Energy-Efficient Buildings Facilitated by Microgrid," *IEEE Trans. Smart Grid*, vol. 1, no. 3, pp. 243–252, Dec. 2010..

[4] X. Li and J. Wen, "Review of building energy modeling for control and operation," *Renew. Sustain. Energy Rev.*, vol. 37, pp. 517–537, Sep. 2014.

[5] X. Li and J. Wen, "Building energy consumption on-line forecasting using physics based system identification," *Energy Build.*, vol. 82, pp. 1–12, Oct. 2014.

[6] J. Široký, F. Oldewurtel, J. Cigler, and S. Prívara, "Experimental analysis of model predictive control for an energy efficient building heating system," *Appl. Energy*, vol. 88, no. 9, pp. 3079–3087, Sep. 2011.

[7] J. Ma, S. J. Qin, B. Li, and T. Salsbury, "Economic model predictive control for building energy systems," in *ISGT 2011*, 2011, pp. 1–6.

[8] F. Oldewurtel, A. Parisio, C. N. Jones, D. Gyalistras, M. Gwerder, V. Stauch, B. Lehmann, and M. Morari, "Use of model predictive control and weather forecasts for energy efficient building climate control," *Energy Build.*, vol. 45, pp. 15–27, Feb. 2012.

[9] B. Yuce, Y. Rezgui, and M. Mourshed, "ANN–GA smart appliance scheduling for optimised energy management in the domestic sector," *Energy Build.*, vol. 111, pp. 311–325, Jan. 2016.

[10] S. A. Kalogirou, "Artificial Neural Networks and Genetic Algorithms in Energy Applications in Buildings," *Adv. Build. Energy Res.*, vol. 3, no. 1, pp. 83–119, Jan. 2009.

[11] B. Yuce, H. Li, Y. Rezgui, I. Petri, B. Jayan, and C. Yang, "Utilizing artificial neural network to predict energy consumption and thermal comfort level: An indoor swimming pool case study," *Energy Build.*, vol. 80, pp. 45–56, Sep. 2014.

[12] W. Kim, Y. Jeon, and Y. Kim, "Simulation-based optimization of an integrated daylighting and HVAC system using the design of experiments method," *Appl. Energy*, vol. 162, pp. 666–674, Jan. 2016.

[13] A.-T. Nguyen, S. Reiter, and P. Rigo, "A review on simulation-based optimization methods applied to building performance analysis," *Appl. Energy*, vol. 113, pp. 1043–1058, Jan. 2014.